Kahan's method on quadratic lifts: modified invariants, Liouville measures and integrable maps in every dimension

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Abstract

Kahan's method discretises quadratic vector fields by birational maps that cost one linear solve per step and often inherit measures, integrals and integrability. Quadratic lifts (auxiliary variables such as $y = x_i x_j$ or $u = \mathrm{e}^q$) carry the method beyond quadratic problems, but constraints and invariants need not survive. We show which do. (i) A *block theorem*: variables that move along fixed affine flows at affine speeds keep modified versions of the quadratic invariants of those flows when their second-derivative operators close and commute; the flows themselves need not commute. It gives $I - \frac{h^2}{8} F^\top \nabla^2 I \, F$ when the curvature of I lies in affinely moving variables. (ii) Constraints with affinely moving factors are protected; on the invariant *shifted sheet* the lifted method is a symmetric one-step method for the original system. For quartic potentials in one degree of freedom it preserves an explicit area form. (iii) For potentials built from exponentials and single harmonics, with any mass matrix, the lifted map preserves the Liouville measure and a modified energy. (iv) For isotropic quartic oscillators in $\mathbb{R}^n$ the lifted map is integrable for every n, with all integrals and the measure in closed form. Its radial part is a QRT map, solved explicitly by elliptic functions, and the planar case stays integrable in a magnetic field. This gives an $O(n)$-cost integrator, stable far beyond the Störmer–Verlet limit, whose single failure band the theory predicts. All proved integrable cases stem from linear symmetries and one degree of freedom; degree growth suggests that hidden symmetries do not survive lifting.

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1 Introduction

Let $\dot{z} = F(z)$ be a quadratic vector field on $\mathbb{R}^N$, $F(z) = Q(z,z) + Bz + b$ with Q symmetric and bilinear. Kahan's method [1, 2] replaces it by the map $z \mapsto z'$ defined by

$$\frac{z' - z}{h} = Q(z, z') + \tfrac{1}{2} B(z + z') + b. \tag{1}$$

The new point enters linearly, so a step costs one linear solve. The method is symmetric and, when applied to Hamiltonian quadratic vector fields (cubic Hamiltonians) with constant Poisson structure, it preserves a modified energy and an invariant measure [7]. Applied to integrable systems it produces integrable maps far more often than one has a right to expect: the Euler and Lagrange tops [3, 4], the Clebsch and Kirchhoff systems and many others [5, 6, 8, 12, 13], planar cubic Hamiltonian systems [13, 15], and cases where integrability can be restored by small

adjustments of the coefficients [14]. Kahan's method is therefore one of the few places where geometric numerical integration [28] and the theory of discrete integrable systems [29, 30] meet.

The method is defined only for quadratic vector fields. Polarisation extends it to higher degree at the price of multistep schemes [9], and Hone and Quispel construct explicit analogues for certain higher order equations [19]. Another route is a *quadratic lift*. Introduce auxiliary variables, for instance $y_{ij} = x_i x_j$, $u = \mathrm{e}^{\alpha \cdot q}$ or $(c, s) = (\cos\theta, \sin\theta)$, so that the original system becomes the restriction of a quadratic system in more variables to an invariant "sheet". Every polynomial vector field admits such a lift [21], and lifts are widely used in model reduction [22, 23, 24]. In geometric integration, auxiliary variables make energy-stable schemes linear (the IEQ and SAV approaches [25, 26]), and multiple quadratic auxiliary variables allow the midpoint rule to preserve several invariants at once [27]. Combining a quadratic lift with Kahan's method gives a linearly implicit one-step integrator for any polynomial ODE, and for many non-polynomial ones.

The price is that Kahan's map lives in the lifted space. It need not preserve the sheet, and the integrals and measures of the original system need not survive. This paper asks which structures do survive, and whether lifting can produce new integrable maps. Our main results are the following.

(i) **A block theorem** (Theorem 3.2). If a block of variables moves as $\dot{w} = \sum_k A_k(z) L_k(w)$, with affine *speeds* A_k and fixed affine vector fields L_k, then Kahan's map conserves explicitly modified versions of all quadratic invariants of the L_k, provided the second-derivative operators of the L_k close on these invariants and commute. The fields L_k themselves need not commute. The theorem contains the two-variable theorem of [10], gives modified invariants with squared denominators for resonant blocks (Example 3.6), and yields $I_h = I - \frac{h^2}{8} F^\top \nabla^2 I \, F$ whenever the curvature of a quadratic integral I lies in variables that move affinely (Corollary 3.4).

(ii) **Lift constraints and the shifted sheet** (Section 4). We classify the lift constraints $y - x_i x_j$ that are exact integrals (Proposition 4.1). Those whose factors move affinely are always protected, as $y - x_i x_j + \frac{h^2}{4} \dot{x}_i \dot{x}_j$ (Theorem 4.2). On the invariant "shifted sheet" that these modified constraints define, the lifted method is a symmetric second-order one-step method on the original variables (Theorem 4.4).

(iii) **Invariant measures** (Theorem 5.1 and Proposition 5.3). For every Hamiltonian $\frac{1}{2} p^\top G p + \sum_k c_k \mathrm{e}^{\alpha_k \cdot q} + \sum_j g_j \cos(\beta_j \cdot q)$, in any dimension and with any constant mass matrix, Kahan's method in the exponential and harmonic variables preserves the canonical Liouville measure and a modified energy. This covers Toda chains, rotor and XY chains and coupled pendula. For every potential of degree at most four in one degree of freedom, the lifted map preserves an explicit measure, so the method on the shifted sheet preserves an area form even when it is not integrable (Corollary 5.4).

(iv) **An integrable family in every dimension** (Theorem 6.1 and Corollary 6.2). For $H = \frac{1}{2}|p|^2 + U(|q|^2)$ with U quadratic and $q, p \in \mathbb{R}^n$, the lift $y = |q|^2$ gives a Kahan map that conserves modified angular momenta, a modified energy and an explicit measure, for every n. Its compact regular level sets are invariant 2-tori on which it acts as a skew rotation. Its radial part is a QRT map: the lifted variable y_k is an elliptic function sampled at equal steps, which gives an explicit solution (Theorem 6.5 and Corollary 6.6). The map can be evaluated in closed form at $O(n)$ cost (Remark 6.4), and bounded orbits are guaranteed under an explicit condition (Proposition 6.3). In a scan over step sizes it stays stable far beyond the Störmer–Verlet limit, with a single failure band where this condition fails, whereas a splitting method that integrates the stiff part exactly resonates near every multiple of $\pi/2$ (Section 8).

(v) **Which integrable systems survive lifting** (Section 7). We prove integrability for every lifted map in our survey whose degree growth is polynomial (Proposition 7.1). The

cases are one degree of freedom, the isotropic family, single exponentials and harmonics, products of these up to affine changes of variables, and the planar isotropic oscillator in a magnetic field. For the last, whose integrals lie outside the reach of the block theorem, we give the integrals and the measure in closed form, together with a discrete Larmor identity (Proposition 7.2). Exact degree-growth computations on nineteen lifted maps indicate that integrability resting on hidden symmetries is lost (Conjecture 7.3).

Relation to other work. A companion manuscript [20] studies Kahan's method for Lie–Poisson systems through a Cayley form of the map. Its "midpoint theorem" covers the single-speed case of our block theorem and the commuting cases of our measure results. Our block theorem does not need commuting coefficient matrices, which is what the isotropic family requires (Remark 3.8). Conversely, [20] proves, with computer assistance, that every three-dimensional quadratic Nambu system has a Kahan-invariant measure; for the Nambu lifts of quartic potentials we give this measure in closed form (Proposition 5.3). Integrable Kahan maps in arbitrary dimension are known for Lotka–Volterra systems [40, 41] and for almost abelian Lie–Poisson systems [20]; the family of Section 6 is of a different kind, coming from a mechanical system with a non-quadratic potential through a lift. Integrable discretisations of central-force and quartic-potential problems have been constructed by other means [31, 32, 30], and discrete-gradient methods preserve energy and angular momentum [33]. What is new in Section 6 is that integrability arises from a general-purpose, linearly implicit integrator, with all modified integrals and the measure in closed form and in every dimension. Kahan discretisations of planar quadratic Hamiltonian systems are known to give QRT-type maps [13, 15]; here a QRT map appears as the radial part of a map in dimension $2n+1$.

Outline. Section 2 collects facts about Kahan's method and defines quadratic lifts. Section 3 proves the block theorem and its corollaries. Section 4 treats lift constraints and the shifted sheet, Section 5 invariant measures, and Section 6 the isotropic family and its radial dynamics. Section 7 proves the integrable cases and surveys the others, Section 8 gives numerical experiments, and Section 9 lists open problems. Appendix A describes the computer-algebra verifications, and Appendix B the exact searches.

2 Preliminaries

2.1 Kahan's method

Throughout, $F(z) = Q(z,z) + Bz + b$ is a quadratic vector field on $\mathbb{R}^N$, $F'(z) = 2Q(z,\cdot) + B$ is its Jacobian matrix, and $\Phi_h\colon z \mapsto z'$ is Kahan's map (1). We collect standard facts, see [7, 8].

Proposition 2.1. *(a)* $\Phi_h(z) = z + h\big(\mathrm{Id} - \frac{h}{2}F'(z)\big)^{-1}F(z)$*;* Φ_h *is birational and* $\Phi_h^{-1} = \Phi_{-h}$*.*

(b) If a, b *are affine functions, Kahan's method discretises the quadratic term* $a(z)b(z)$ *as* $\frac{1}{2}[a(z)b(z') + a(z')b(z)]$*.*

(c) Kahan's method commutes with affine changes of variables: if $z = Tw + c$ *with* T *invertible and* $G(w) = T^{-1}F(Tw+c)$*, then the Kahan map of* G *is* $w \mapsto T^{-1}(\Phi_h(Tw+c) - c)$*.*

(d) $\det D\Phi_h(z) = \det\big(\mathrm{Id} + \frac{h}{2}F'(z')\big)/\det\big(\mathrm{Id} - \frac{h}{2}F'(z)\big)$*.*

(e) Every linear first integral of F *is a first integral of* Φ_h*.*

Definition 2.2. Let I be a function on $\mathbb{R}^N$. A *modified integral* of I is a function I_h, rational in z and smooth in h near $h = 0$, with $I_0 = I$ and $I_h \circ \Phi_h = I_h$. We call I *protected* if it has a modified integral. A measure $\mu = \mathrm{d}z/P(z)$ with P rational is *preserved* if

$$P(\Phi_h(z)) = \det D\Phi_h(z)\, P(z), \tag{2}$$

i.e. P is a discrete Darboux function whose cofactor is the Jacobian determinant [11].

Lemma 2.3. *A protected function is a first integral of F, and the protected functions form a vector space.*

Proof. $0 = I_h(\Phi_h(z)) - I_h(z) = h\,\nabla I(z)\cdot F(z) + O(h^2)$. The second claim holds because modified integrals can be added. □

2.2 Quadratic lifts

Definition 2.4. Let $\dot{x} = f(x)$ be an ODE on $\mathbb{R}^n$. A *quadratic lift* consists of a map $\psi\colon \mathbb{R}^n \to \mathbb{R}^m$ and a quadratic vector field F on $\mathbb{R}^{n+m}$ with $F(x,\psi(x)) = \big(f(x), D\psi(x)f(x)\big)$. The *sheet* $\Sigma = \{y = \psi(x)\}$ is invariant under the flow of F, and the flow on Σ projects to the flow of f. The components of $c = y - \psi(x)$ are the *constraints*. A constraint is *exact* if it is a first integral of F on all of $\mathbb{R}^{n+m}$, and *conditional* if its derivative vanishes only on Σ. The *lifted Kahan method* is Kahan's method applied to F.

The lift is not unique: on Σ the monomial q^2 may be written as q^2 or as y, and more generally any multiple of the constraints may be added to F. By Lemma 2.3, conditional constraints are never protected.

Example 2.5 (the quartic oscillator)**.** For $H = \frac{1}{2}p^2 + \frac{1}{4}q^4$ and $y = q^2$, the lifted field is $\dot{q} = p$, $\dot{p} = -qy$, $\dot{y} = 2qp$. The constraint $c = y - q^2$ is exact, and so is the lifted energy $\tilde{H} = \frac{1}{2}p^2 + \frac{1}{4}y^2$. Kahan's map conserves $c_h = y - q^2 + \frac{h^2}{4}p^2$ and $\tilde{H}_h = \tilde{H}/(1 + \frac{1}{2}h^2q^2)$, and it preserves $\mathrm{d}q\,\mathrm{d}p\,\mathrm{d}y/(1 + \frac{1}{2}h^2q^2)$. Both integrals follow from Section 3.

3 The block theorem

3.1 Statement and proof

Let $P\colon \mathbb{R}^N \to \mathbb{R}^m$ be linear and $w = Pz$ a block of linear coordinates. Suppose that

$$PF(z) = \sum_{k=1}^{r} A_k(z)\, L_k(Pz), \tag{3}$$

where the *speeds* $A_k\colon \mathbb{R}^N \to \mathbb{R}$ are affine and the $L_k(w) = L_k w + \ell_k$ are fixed affine vector fields on $\mathbb{R}^m$. For a function I of degree at most two on $\mathbb{R}^m$, with Hessian $\nabla^2 I$, define

$$(\mathcal{T}_{kl} I)(w) = L_k(w)^{\top}\, \nabla^2 I\, L_l(w).$$

Lemma 3.1 (crossed identity)**.** *Assume (3), let $z' = \Phi_h(z)$, $w = Pz$, $w' = Pz'$, $a_k = \frac{h}{2}A_k(z)$ and $a'_k = \frac{h}{2}A_k(z')$. Then*

$$w' - \sum_k a_k L_k(w') = w + \sum_k a'_k L_k(w), \tag{4}$$

and every polynomial I of degree at most two on $\mathbb{R}^m$ that is a first integral of every L_k satisfies

$$I(w') + \frac{1}{2}\sum_{k,l} a_k a_l\, (\mathcal{T}_{kl} I)(w') = I(w) + \frac{1}{2}\sum_{k,l} a'_k a'_l\, (\mathcal{T}_{kl} I)(w). \tag{5}$$

Proof. By Proposition 2.1(b), applied to each product in (3), the w-components of Kahan's equations read $w' - w = \frac{h}{2}\sum_k \big[A_k(z)L_k(w') + A_k(z')L_k(w)\big]$, which is (4). Since I is quadratic, Taylor's formula is exact, and since $\nabla I \cdot L_k = 0$, $I\big(w \pm \sum_k a_k L_k(w)\big) = I(w) + \frac{1}{2}\sum_{k,l} a_k a_l (\mathcal{T}_{kl} I)(w)$. Applying I to both sides of (4) gives (5). □

Theorem 3.2 (block theorem)**.** *Let V be a finite-dimensional space of polynomials of degree at most two on $\mathbb{R}^m$, each a first integral of every L_k, and assume that $\mathcal{T}_{kl}V \subset V$ for all k, l. Choose a basis $\mathbf{I} = (I_1, \dots, I_d)$ of V and let T_{kl} be the matrices with $\mathcal{T}_{kl} I_i = \sum_j (T_{kl})_{ij} I_j$. If the matrices T_{kl} commute pairwise, then the vector of functions*

$$\mathbf{I}_h(z) = \mathbf{M}_h(z)^{-1}\, \mathbf{I}(Pz), \qquad \mathbf{M}_h(z) = \mathrm{Id} + \frac{h^2}{8} \sum_{k,l=1}^{r} A_k(z) A_l(z)\, T_{kl}, \tag{6}$$

is conserved by Kahan's map wherever $\mathbf{M}_h$ is invertible.

Proof. Apply Lemma 3.1 to each basis element. With $\mathbf{M}(a) = \mathrm{Id} + \frac{1}{2}\sum_{k,l} a_k a_l T_{kl}$ we obtain $\mathbf{M}(a)\mathbf{I}(w') = \mathbf{M}(a')\mathbf{I}(w)$. The matrices $\mathbf{M}(a)$ and $\mathbf{M}(a')$ are polynomials in the commuting T_{kl}, so they commute, and therefore $\mathbf{M}(a')^{-1}\mathbf{I}(w') = \mathbf{M}(a)^{-1}\mathbf{I}(w)$. Since $\mathbf{M}(a) = \mathbf{M}_h(z)$ and $\mathbf{M}(a') = \mathbf{M}_h(z')$, this is the claim. □

The "speeds" cross over in (4): the speed at the old point multiplies the new point and vice versa. If all $a_k = a'_k$, (4) is a Cayley transform and conserves every element of V exactly. The rescaling $\mathbf{M}_h$ is the exact price of sampling the speeds at different ends of the step.

3.2 Consequences

Corollary 3.3 (two variables, [10])**.** *Let $w = Pz \in \mathbb{R}^2$ and $I(w) = \frac{1}{2}a_1 w_1^2 + a_2 w_1 w_2 + \frac{1}{2}a_3 w_2^2 + a_4 w_1 + a_5 w_2$. If $\dot{w} = A(z) J \nabla I(w)$ with A affine and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, then Kahan's map conserves*

$$I_h = \frac{I + \frac{1}{8}h^2 D_2 A^2}{1 + \frac{1}{4}h^2 D_1 A^2}, \qquad D_1 = a_1 a_3 - a_2^2, \quad D_2 = 2a_2 a_4 a_5 - a_3 a_4^2 - a_1 a_5^2.$$

Proof. Take $V = \mathrm{span}\{I, 1\}$ and $L(w) = J\nabla I(w)$. With $S = \nabla^2 I$ and $s = (a_4, a_5)^\top$ we have $J^\top S J = \mathrm{adj}\, S$, hence $\mathcal{T} I = (Sw + s)^\top \mathrm{adj}(S)(Sw + s) = 2D_1 I - D_2$ and $\mathcal{T} 1 = 0$. Inverting the 2×2 triangular matrix $\mathbf{M}_h$ gives the formula. □

Corollary 3.4 (affine factors)**.** *Let $I(z) = \frac{1}{2} z^\top S z + s^\top z$ be a first integral of F with $S\, Q(u, v) = 0$ for all u, v, i.e. every linear form in the range of S has an affine time derivative. Then*

$$I_h = I - \frac{h^2}{8} F(z)^\top S\, F(z)$$

is a modified integral.

Proof. Write $S = R^\top \Lambda R$ with R of full row rank and Λ symmetric and invertible, and let $\phi = Rz$ and $\chi = s^\top z$. Since $RQ = 0$, $\dot{\phi} = \alpha(z)$ with α affine, and $\nabla I \cdot F = 0$ gives $\dot{\chi} = -\phi^\top \Lambda \alpha(z)$. The block $w = (\phi, \chi)$ therefore moves as $\dot{w} = \sum_j \alpha_j(z) L_j(w)$ with $L_j(w) = (e_j, -(\Lambda\phi)_j)$. The function $\tilde{I}(w) = \frac{1}{2}\phi^\top \Lambda \phi + \chi$ equals I, is invariant under every L_j, and satisfies $\mathcal{T}_{jk}\tilde{I} = \Lambda_{jk}$ and $\mathcal{T}_{jk} 1 = 0$. All T_{jk} are multiples of one nilpotent matrix, so Theorem 3.2 applies and gives $I_h = I - \frac{h^2}{8}\alpha^\top \Lambda \alpha = I - \frac{h^2}{8} F^\top S F$. □

Example 3.5. For $\dot{x} = y$, $\dot{y} = z$, $\dot{z} = 12xy$ the integral $I = z - 6x^2$ has $S = -12 e_x e_x^\top$, and $\dot{x}$ is linear. Corollary 3.4 gives $I_h = z - 6x^2 + \frac{3}{2}h^2 y^2$, an invariant found by direct computation in [8]. In Example 2.5, the corollary gives c_h.

Example 3.6 (resonant blocks)**.** Let $\mathbb{J}$ be a 2×2 rotation by a right angle and let a block $(u, v) \in \mathbb{R}^2 \times \mathbb{R}^2$ move as $\dot{u} = A(z)(\omega \mathbb{J} u + v)$, $\dot{v} = A(z)\, \omega \mathbb{J} v$. The quadratic invariants of the fixed (Jordan-block) flow include $N = |v|^2$ and $G = u^\top \mathbb{J} v$, and $\mathcal{T} N = 2\omega^2 N$, $\mathcal{T} G = 2\omega^2 G - 2\omega N$. With $\beta = \frac{1}{4}h^2 \omega^2 A(z)^2$, Theorem 3.2 gives

$$N_h = \frac{N}{1 + \beta}, \qquad G_h = \frac{G}{1 + \beta} + \frac{\beta N/\omega}{(1 + \beta)^2}.$$

The squared denominator is not produced by Corollaries 3.3 and 3.4: in normal-mode coordinates G is not a sum of two-variable integrals, and its Hessian does not annihilate the quadratic part.

Example 3.7 (non-commuting fields: isotropic angular momenta)**.** For $H = \frac{1}{2}|p|^2 + U(|q|^2)$ with U quadratic, lifted by $y = |q|^2$, the block $w = (q, p) \in \mathbb{R}^{2n}$ moves as $\dot{w} = 1 \cdot L_1(w) + 2U'(y) \cdot L_2(w)$ with $L_1(q, p) = (p, 0)$ and $L_2(q, p) = (0, -q)$. These shear fields do not commute; together with $[L_1, L_2]$ they span $\mathfrak{sl}(2)$. The angular momenta $M_{ab} = q_a p_b - q_b p_a$ are invariant under both, and $\mathcal{T}_{11} M_{ab} = \mathcal{T}_{22} M_{ab} = 0$, $\mathcal{T}_{12} M_{ab} = \mathcal{T}_{21} M_{ab} = M_{ab}$. Theorem 3.2 gives the modified angular momenta

$$M_{ab,h} = \frac{M_{ab}}{1 + \frac{1}{2}h^2 U'(y)}.$$

Remark 3.8. When all coefficient matrices of the Cayley form of Kahan's map commute, [20] shows that z and z' share a "midpoint" at which every compatible quadratic form is conserved. This covers Corollary 3.3, Corollary 3.4 and the single-speed case of Theorem 3.2. In Example 3.7 the coefficient matrices do not commute: their commutator is a non-zero diagonal matrix. Theorem 3.2 only needs the operators $\mathcal{T}_{kl}$ to commute *on* V. For the generalised Suslov systems $\dot{x} = \ell(x) J_0 \mathsf{H} x$ of [8], Theorem 3.2 gives one modified integral $I_k/(1 + \frac{1}{4}h^2 \ell(x)^2 \omega_k^2)$ for each normal-mode action I_k with frequency ω_k.

4 Lift constraints and the shifted sheet

4.1 Which constraints are exact, and which are protected

Proposition 4.1. *Let $c = y - x_i x_j$ be a first integral of a quadratic field F, where y, x_i, x_j are coordinates, and write $F_i = Q_i + (\text{affine})$, $F_j = Q_j + (\text{affine})$ with quadratic forms Q_i, Q_j. Then $x_j Q_i + x_i Q_j = 0$. Consequently:*

(a) if $i = j$, then $Q_i = 0$;

(b) if $i \neq j$, then either $Q_i = Q_j = 0$, or $Q_i = x_i R$ and $Q_j = -x_j R$ for a non-zero linear form R.

Proof. $\nabla c \cdot F = F_y - x_j F_i - x_i F_j = 0$, and F_y is quadratic, so the cubic part $x_j Q_i + x_i Q_j$ vanishes. For $i = j$ this gives $Q_i = 0$. For $i \neq j$ the coordinates x_i and x_j are coprime, so x_i divides Q_i; writing $Q_i = x_i R$ gives $Q_j = -x_j R$. □

In case (b) with $R \neq 0$ the two factors are stretched and squeezed at the same rate, like $(q, p) \mapsto (\mathrm{e}^{\tau} q, \mathrm{e}^{-\tau} p)$. We call this *opposite scaling*.

Theorem 4.2. *If $c = y - x_i x_j$ is a first integral and $\dot{x}_i = F_i$, $\dot{x}_j = F_j$ are affine (cases (a) and (b) with $Q_i = Q_j = 0$), then*

$$c_h = y - x_i x_j + \frac{h^2}{4} F_i(z) F_j(z)$$

is a modified integral.

Proof. Apply Corollary 3.4: $S = -(e_i e_j^{\top} + e_j e_i^{\top})$ annihilates Q exactly when $Q_i = Q_j = 0$, and $F^{\top} S F = -2 F_i F_j$. □

Corollary 4.3 (mechanical systems with quartic potentials)**.** *Let $H = \frac{1}{2} p^{\top} G p + V(q)$ with G constant and V a polynomial of degree at most four, and lift with $Y = (y_{ij}) = q q^{\top}$. For every lifted field with $\dot{q} = Gp$ and $\dot{Y} = \dot{q} q^{\top} + q \dot{q}^{\top}$, Kahan's map conserves the matrix $Y - q q^{\top} + \frac{h^2}{4} (Gp)(Gp)^{\top}$.*

In case (b) with $R \neq 0$, Theorem 3.2 does not apply. The block (x_i, x_j, y) moves along the scaling field $(x_i, -x_j, 0)$ and the translation fields $(1, 0, x_j)$, $(0, 1, x_i)$, and the second derivative of c along the scaling field, $2 x_i x_j$, is not an invariant of the translations, so closure fails. Such a constraint can still be protected through a *partner*, i.e. by lying in the vector space of Lemma 2.3. Proposition 4.6 gives an example.

4.2 Back-substitution: the shifted sheet

Theorem 4.4. *Let F be a quadratic lift of $\dot{x} = f(x)$ whose constraints $c = y - \psi(x)$ all have modified integrals of the form $c_h = c + h^2 r_h(x, y)$ with r_h smooth. For h small, the* shifted sheet *$\Sigma_h = \{c_h = 0\}$ is locally the graph of $y = \psi_h(x) = \psi(x) + O(h^2)$, it is invariant under Φ_h, and the induced map $\varphi_h \colon x \mapsto x'$ is a one-step method of order two for $\dot{x} = f(x)$. If $c_{-h} = c_h$, then φ_h is symmetric, $\varphi_h^{-1} = \varphi_{-h}$. One step costs one linear solve of size $n + m$.*

Proof. The implicit function theorem gives ψ_h, since $\partial c_h/\partial y = \mathrm{Id} + O(h^2)$. Invariance follows from $c_h \circ \Phi_h = c_h$. Since $\Phi_h(z) = z + O(h)$ and $\Phi_h(z + \delta) = \Phi_h(z) + \delta + O(h\delta)$, starting at $(x, \psi_h(x))$ instead of $(x, \psi(x))$ changes the x-component of the step by $O(h^3)$. On Σ the exact flow of F projects to that of f, and Kahan's method has order two, so φ_h has order two. If $c_{-h} = c_h$ then $\Sigma_{-h} = \Sigma_h$, and $\Phi_h^{-1} = \Phi_{-h}$ gives $\varphi_h^{-1} = \varphi_{-h}$. □

Example 4.5. For the quartic oscillator, $\Sigma_h = \{y = q^2 - \frac{h^2}{4}p^2\}$ and substituting into Kahan's equations gives

$$q' = q + \tfrac{h}{2}(p + p'), \qquad p' = p - \tfrac{h}{2}\, qq'(q + q') + \tfrac{h^3}{8}\big(qp'^2 + q'p^2\big).$$

This symmetric map of the plane is *nonlinearly* implicit in (q, p), yet the lifted form computes it with one linear solve. For mechanical lifts (Corollary 4.3) the shifted sheet is explicit, $Y = qq^\top - \frac{h^2}{4}(Gp)(Gp)^\top$.

4.3 One degree of freedom: protection through a partner

Let $x = (q, p) \in \mathbb{R}^2$ and consider $H = U(\varphi(x)) + \kappa(x)$ with $\varphi = \frac{1}{2}x^\top \Phi x$, $\kappa = \frac{1}{2}x^\top K x$ and $U(y) = \frac{1}{2}u_2 y^2 + u_1 y$, $u_2 \neq 0$. Lift with $y = \varphi(x)$ and $\tilde{H} = U(y) + \kappa(x)$. The lifted field is the Nambu field $\dot{z} = \nabla\tilde{H} \times \nabla c$ on $z = (q, p, y)$, with $c = y - \varphi(x)$. This covers all even quartic oscillators ($\varphi = q^2$) and, for $\varphi = qp$, an opposite-scaling constraint.

Proposition 4.6 (essentially [10, Theorem 2])**.** *Let $\det\Phi \neq 0$, and let $\lambda_1 \neq \lambda_2$ be the roots of $\det(K - \lambda\Phi) = 0$ (real, or complex conjugate). Then each pencil member $I_\lambda = \tilde{H} + \lambda c$ is a two-variable integral, $K - \lambda\Phi = \mu_\lambda \ell_\lambda \ell_\lambda^\top$, and with $A_\lambda = \ell_\lambda^\top J \Phi x$,*

$$I_{\lambda,h} = \frac{I_\lambda - \frac{1}{8}h^2\mu_\lambda(u_1 + \lambda)^2 A_\lambda^2}{1 + \frac{1}{4}h^2\mu_\lambda u_2 A_\lambda^2}.$$

Hence $c_h = (I_{\lambda_1,h} - I_{\lambda_2,h})/(\lambda_1 - \lambda_2)$ and $\tilde{H}_h$ are conserved, the measure $\mathrm{d}z/(D_{\lambda_1} D_{\lambda_2})$ with $D_\lambda = 1 + \frac{1}{4}h^2\mu_\lambda u_2 A_\lambda^2$ is preserved, and the lifted Kahan map is integrable.

Proof sketch. In the coordinates $(\ell_{\lambda_1}^\top x, y, \ell_{\lambda_2}^\top x)$ both pencil members are two-variable functions sharing y, and $\nabla\tilde{H} \times \nabla c$ is a constant multiple of $\nabla I_{\lambda_1} \times \nabla I_{\lambda_2}$. This is the integrable class of [10, Theorem 2]; the formula for $I_{\lambda,h}$ is Corollary 3.3. □

If $\det\Phi = 0$, one root lies at infinity: φ is then the square of a linear form, c_h is given by Theorem 4.2, $I_{\lambda_1,h}$ by the formula above for the finite root λ_1, and the preserved measure is $\mathrm{d}z/D_{\lambda_1}$. The even quartic oscillators ($\varphi = q^2$) are of this kind.

Remark 4.7 (limits of protection). Protection by Corollary 3.4 reaches one level deep: the *factors* of the constraint must move affinely. In mechanical systems these are the positions, so lifts by products of positions are protected; they represent potentials of degree at most four. A sextic potential needs products of lifted variables, whose velocities are quadratic, and the corresponding constraints are conditional. Rewriting a product by polarisation, $x_i x_j = \frac{1}{2}[(x_i + x_j)^2 - x_i^2 - x_j^2]$, does not help: the condition $SQ = 0$ is invariant under linear changes of variables (Proposition 2.1(c)). Exponentials and single harmonics escape the one-level rule, because their derivatives stay in the span of the lifted variables (Section 5).

5 Invariant measures

5.1 Exponential and trigonometric potentials: the Liouville measure

Consider, on $\mathbb{R}^n \times \mathbb{R}^n$,

$$H = \tfrac{1}{2} p^\top G p + \sum_{k=1}^{K} c_k \, \mathrm{e}^{\alpha_k \cdot q} + \sum_{j=1}^{J} g_j \cos(\beta_j \cdot q), \tag{7}$$

with a constant matrix G and arbitrary vectors $\alpha_k, \beta_j \in \mathbb{R}^n$ (dependent or not). With $u_k = \mathrm{e}^{\alpha_k \cdot q}$ and $(C_j, S_j) = (\cos \beta_j \cdot q, \sin \beta_j \cdot q)$, drop q and consider the quadratic system

$$\dot p = -\sum_k c_k u_k \alpha_k + \sum_j g_j S_j \beta_j, \quad \dot u_k = (\alpha_k^\top G p)\, u_k, \quad \dot C_j = -(\beta_j^\top G p)\, S_j, \quad \dot S_j = (\beta_j^\top G p)\, C_j. \tag{8}$$

Theorem 5.1. *Kahan's map of (8) preserves the measure*

$$\mathrm{d}p \prod_{k=1}^{K} \frac{\mathrm{d}u_k}{|u_k|} \prod_{j=1}^{J} \frac{\mathrm{d}C_j \, \mathrm{d}S_j}{C_j^2 + S_j^2}$$

and conserves each modified circle $(C_j^2 + S_j^2)/(1 + \frac{1}{4}h^2(\beta_j^\top G p)^2)$, for any matrix G. If G is symmetric it also conserves $E_h = \frac{1}{2} p^\top G p + \sum_k c_k u_k + \sum_j g_j C_j - \frac{h^2}{8} \dot p^\top G \dot p$, with $\dot p$ given by (8).

Proof. Write $a_k = \alpha_k^\top G p$, $b_j = \beta_j^\top G p$, $\zeta_j = C_j + \mathrm{i} S_j$, and primes for the new values. By Proposition 2.1(b) Kahan's equations are

$$u_k'\big(1 - \tfrac{h}{2} a_k\big) = u_k\big(1 + \tfrac{h}{2} a_k'\big), \qquad \zeta_j'\big(1 - \tfrac{\mathrm{i}h}{2} b_j\big) = \zeta_j\big(1 + \tfrac{\mathrm{i}h}{2} b_j'\big),$$

and $p' - p = \frac{h}{2}\big[-\sum_k c_k \alpha_k (u_k + u_k') + \sum_j g_j \beta_j (S_j + S_j')\big]$. Define intermediate variables $\hat u_k = u_k/(1 - \frac{h}{2} a_k)$ and $\hat\zeta_j = \zeta_j/(1 - \frac{\mathrm{i}h}{2} b_j)$. Then $u_k + u_k' = \hat u_k (2 + \frac{h}{2} \alpha_k^\top G \delta)$ and $S_j + S_j' = 2\hat S_j + \frac{h}{2} \hat C_j \beta_j^\top G \delta$, where $\delta = p' - p$, so the momentum equation becomes

$$\Big(\mathrm{Id} + \frac{h^2}{4}\Big[\sum_k c_k \hat u_k \alpha_k \alpha_k^\top - \sum_j g_j \hat C_j \beta_j \beta_j^\top\Big] G\Big)\delta = -h \sum_k c_k \hat u_k \alpha_k + h \sum_j g_j \hat S_j \beta_j.$$

Thus δ depends only on $(\hat u, \hat\zeta)$. In the coordinates p, $x_k = \log|u_k|$, $\rho_j = \log|\zeta_j|$, $\theta_j = \arg \zeta_j$, one step is the composition of three shears: (x, ρ, θ) is shifted by a function of p; then p is shifted by $\delta(\hat u, \hat\zeta)$; then (x, ρ, θ) is shifted by a function of p'. Each shear preserves $\mathrm{d}p \, \mathrm{d}x \, \mathrm{d}\rho \, \mathrm{d}\theta$, which is the stated measure. The energy is Corollary 3.4: $\dot p$ is linear in (u, S), so the Hessian of the lifted energy, which is G on the p-block, annihilates Q. Taking moduli in the ζ_j-equation gives the circles. □

Corollary 5.2. *For the open and periodic Toda chains, rotor and XY chains with nearest-neighbour coupling, and coupled pendula with a non-diagonal mass matrix, the lifted Kahan method is a reversible, volume-preserving, linearly implicit one-step method that conserves a modified energy (and, for Toda, the total momentum).*

The map is not symplectic: for three-particle Toda in the coordinates $(p, \log u)$, the canonical Poisson structure is violated by an amount of order h^3 (about 10^{-2} at $h = 0.2$). Volume preservation by Runge–Kutta methods, of which Kahan's method on quadratic fields is an instance [7], is studied in [47]. The Liouville measure is h-independent, which is unusual for Kahan maps, whose densities typically depend on h. For Lotka–Volterra systems the analogous measure $\mathrm{d}x / \prod_i x_i$ is studied in [40, 41, 20]; Theorem 5.1 concerns the mixed case, with momenta that move affinely and exponential or harmonic variables that move multiplicatively. Volume preservation and reversibility are the two properties required of an integrator in Hybrid Monte Carlo [39], so Theorem 5.1 provides linearly implicit candidates for that purpose.

5.2 One degree of freedom: an area form on the shifted sheet

Write a one-degree-of-freedom potential of degree at most four as $V(q) = a_4q^4 + a_3q^3 + a_2q^2 + a_1q$, and let $\tilde V(q, y) = a_4y^2 + a_3qy + a_2y + a_1q$, so that $\tilde V(q, q^2) = V(q)$. The lift $\dot q = p$, $\dot p = -\tilde V_q - 2q\tilde V_y$, $\dot y = 2qp$ is the Nambu field $\nabla\tilde H \times \nabla c$ on $\mathbb{R}^3$ with $\tilde H = \frac12 p^2 + \tilde V$ and $c = y - q^2$. Any other quadratic $\tilde V$ with $\tilde V(q, q^2) = V(q)$ differs from this one by a multiple of c and gives the same field. Other lifts of the same potential, which differ by multiples of $y - q^2$ in $\dot p$, are in general not Nambu fields. Kahan maps of three-dimensional quadratic systems have a rich structure [42, 8, 43]. The companion manuscript [20] proves, with computer assistance, that Kahan's map of every three-dimensional Nambu field with quadratic H and C preserves a measure $\mathrm{d}z/P$ with P a polynomial of degree at most four. For the Nambu lifts of quartic potentials we give this measure in closed form.

Proposition 5.3. *For every V as above, Kahan's map of the Nambu lift preserves the measure $\mathrm{d}q\,\mathrm{d}p\,\mathrm{d}y/P$, where*

$$P = 3 + h^2T_1 + \tfrac{1}{16}h^4T_2 + \tfrac{1}{16}h^6\,(4a_4q + a_3)^2\,p^2, \qquad T_1 = 4a_4q^2 + 2a_4y + 3a_3q + a_2,$$
$$T_2 = -4\big(2a_4w + a_3q + a_2\big)^2 + 8\big(4a_2a_4 - a_3^2\big)\,w + 8a_1(4a_4q + a_3), \qquad w = 2q^2 - y.$$

On the sheet $y = q^2$ one has $T_1 = \frac12 V''$, $4a_4q + a_3 = \frac16 V'''$ and $T_2 = \frac43 V'V''' - V''^2$.

Proof. Let $\mathcal{M} = \mathrm{Id} - \frac{h}{2}F'(z)$ and $\Delta = \det\mathcal{M}$, so that $z' = z + h\,\mathrm{adj}(\mathcal{M})F(z)/\Delta$ by Proposition 2.1(a). Since P has degree four and F' is affine, multiplying the measure identity $P(z')\,\Delta = \det\big(\mathrm{Id} + \frac{h}{2}F'(z')\big)P(z)$ by Δ^4 turns it into an identity between polynomials in $(q, p, y, h, a_1, \dots, a_4)$. We verified this identity, and the values on the sheet, by computer algebra (Appendix A). □

We found P by the exact search of Appendix B, in an ansatz that is invariant under the two scaling symmetries of the family of fields; within this ansatz the density is unique up to a factor that depends only on h and the coefficients. For $V = \frac14 q^4$ it factorises as $P = \big(1 + \frac12 h^2q^2\big)\big(3 + \frac12 h^2c_h - \frac14 h^4\tilde H_h\big)$, where $1 + \frac12 h^2q^2$ is the density of Example 2.5 and c_h, $\tilde H_h$ are its integrals. Two of the four terms of P are invariants of the Jacobian matrix: $h^2T_1 = \frac12 h^2\sigma_2\big(F'(z)\big)$, where σ_2 is the sum of the principal 2×2 minors, and the last term is $\frac{1}{64}h^6\big(\det F'(z)\big)^2$. This suggests a closed form for general three-dimensional Nambu fields (Section 9).

Corollary 5.4. *For every potential of degree at most four, the back-substituted method of Theorem 4.4 for the Nambu lift preserves the area form $\mathrm{d}q \wedge \mathrm{d}p/P\big(q, p, q^2 - \frac{h^2}{4}p^2\big)$.*

Proof. On the invariant level set $\{c_h = 0\}$ the preserved measure $\mathrm{d}z/P$ induces the invariant form ω with $\mathrm{d}z/P = \mathrm{d}c_h \wedge \omega$. Since $\partial c_h/\partial y = 1$, its pull-back along the graph $y = q^2 - \frac{h^2}{4}p^2$ is $\mathrm{d}q \wedge \mathrm{d}p/P$. □

For a non-symmetric quartic such as $V = \frac14 q^4 + \frac13 q^3 + \frac12 q^2$, no modified energy is conserved and the degree growth is exponential (Section 7). The back-substituted map is then an area-preserving near-identity map that is not integrable. Area preservation is what KAM-type arguments such as Moser's twist theorem [38] require in order to produce invariant curves that confine orbits, and it is consistent with the bounded energy error observed in practice.

5.3 Measures are rare beyond these classes

Exact searches for polynomial densities (Appendix B) find none for the lifted Kahan maps of two-site φ^4 and Fermi–Pasta–Ulam lattices (degree at most six) and of the two-site discrete nonlinear Schrödinger equation (degree at most four). This agrees with the conjecture of [20]

that, for generic quadratic Hamiltonians, Kahan's map of a Lie–Poisson system keeps a measure only for linear systems (two-step nilpotent algebras) and in one degree of freedom; the lifted lattices are affine Lie–Poisson systems on nilpotent algebras of higher step. Among the lifted systems we tested in more than three dimensions, measures came only from symmetry (Section 6) or from exponential and harmonic variables (Theorem 5.1). Other classes have measures for other reasons, for instance Hamiltonian quadratic fields with constant Poisson structure [7] and Lotka–Volterra systems [40, 41].

6 An integrable family in every dimension

Let $n \geq 2$, $U(y) = \frac{1}{2}u_2 y^2 + u_1 y$ with $u_2 \neq 0$, and $H = \frac{1}{2}|p|^2 + U(|q|^2)$ on $\mathbb{R}^n \times \mathbb{R}^n$, a particle in a rotation-invariant anharmonic well. The lift $y = |q|^2$ gives the quadratic field on $\mathbb{R}^{2n+1}$

$$\dot q = p, \qquad \dot p = -2U'(y)\, q, \qquad \dot y = 2\, q \cdot p. \tag{9}$$

Write $Y(z) = 1 + \frac{1}{2}h^2 U'(y)$, $M_{ab} = q_a p_b - q_b p_a$, and $K(z) = \frac{1}{2}|p|^2 + U(y) + \frac{u_1^2}{2u_2} = \frac{1}{2}|p|^2 + \frac{u_2}{2}\big(y + \frac{u_1}{u_2}\big)^2$.

Theorem 6.1. *Kahan's map Φ_h of (9) has the following properties.*

(a) It is $O(n)$-equivariant and (q', p') lies in $\operatorname{span}\{q, p\}$*: explicitly $\binom{q'}{p'} = N\binom{q}{p}$ for a 2×2 matrix $N = N(z, z')$ acting on the pair, with $\det N = Y(z')/Y(z)$. In particular $q \wedge q' = h\, q \wedge p / Y(z)$.*

(b) It conserves

$$c_h = y - |q|^2 + \frac{h^2}{4}|p|^2, \qquad M_{ab,h} = \frac{M_{ab}}{Y},$$
$$E_h = \frac{K + \frac{1}{2}h^2 u_2 |M_h|^2}{D}, \qquad D = 1 + h^2 u_2 |q|^2 - \frac{1}{2}h^4 u_2 K,$$

where $|M_h|^2 = \sum_{a<b} M_{ab,h}^2$.

(c) It preserves the measure $\mathrm{d}z / P_n$ with $P_n = Y^{n-1} D$. More precisely, $\det D\Phi_h(z) = \big(Y(z')/Y(z)\big)^{n-1} D(z')/D(z)$.

The linear term u_1 enters only through $\tilde y = y + u_1/u_2$: in the variables $(q, p, \tilde y)$ the field (9) reads $\dot q = p$, $\dot p = -2u_2 \tilde y q$, $\dot{\tilde y} = 2q \cdot p$, and $Y = 1 + \frac{1}{2}h^2 u_2 \tilde y$, $K = \frac{1}{2}|p|^2 + \frac{1}{2}u_2 \tilde y^2$. Since Kahan's method commutes with this affine change of variables, the case $u_1 \neq 0$ is the case $u_1 = 0$ on another level set of c_h. We write $\kappa = h^2 u_2$ and use primes for values at z'.

Proof. (a) By Proposition 2.1(b) Kahan's equations are

$$q' - q = \tfrac{h}{2}(p + p'), \qquad p' - p = -\alpha q' - \alpha' q, \qquad y' - y = h(q \cdot p' + q' \cdot p), \tag{10}$$

where $\alpha = hU'(y)$ and $\alpha' = hU'(y')$. The first two say $\begin{pmatrix} 1 & -h/2 \\ \alpha & 1 \end{pmatrix}\binom{q'}{p'} = \begin{pmatrix} 1 & h/2 \\ -\alpha' & 1 \end{pmatrix}\binom{q}{p}$, where each entry multiplies vectors in $\mathbb{R}^n$. Hence $N = \frac{1}{Y}\begin{pmatrix} 1 - h\alpha'/2 & h \\ -\alpha - \alpha' & 1 - h\alpha/2 \end{pmatrix}$ and $\det N = (1 + \frac{h}{2}\alpha')/(1 + \frac{h}{2}\alpha) = Y'/Y$. Equivariance holds because (9) is $O(n)$-equivariant (Proposition 2.1(c)).

(b) c_h is Corollary 3.4, since $\dot q = p$ is linear. Since $q' \wedge p' = \det N\, q \wedge p$, we get $M'_{ab}/Y' = M_{ab}/Y$; this is also Example 3.7.

For E_h we may take $u_1 = 0$, so that $\tilde y = y$, by the remark before the proof. Consider the block $w = (p, \tilde y)$. It moves as $\dot w = \sum_k q_k L_k(w)$, with the coordinates q_k as speeds and the fixed linear fields $L_k(w) = (-2u_2 \tilde y\, e_k,\ 2p_k)$, and $K(w)$ is invariant under every L_k. The closure condition of Theorem 3.2 fails, but Lemma 3.1 still applies. With $a = \frac{h}{2}q$ and $a' = \frac{h}{2}q'$ it gives

$K(w') + \frac{1}{2}\sum_{k,l} a_k a_l (\mathcal{T}_{kl}K)(w') = K(w) + \frac{1}{2}\sum_{k,l} a'_k a'_l (\mathcal{T}_{kl}K)(w)$, and here $\sum_{k,l} x_k x_l (\mathcal{T}_{kl}K)(w) = 4u_2\big(u_2\tilde{y}^2|x|^2 + (x\cdot p)^2\big) = 4u_2\big(2K|x|^2 - |x\wedge p|^2\big)$ by Lagrange's identity. Hence

$$K'\big(1+\kappa|q|^2\big) - \tfrac{1}{2}\kappa|q\wedge p'|^2 = K\big(1+\kappa|q'|^2\big) - \tfrac{1}{2}\kappa|q'\wedge p|^2. \tag{11}$$

By (a) and the first equation in (10), $q\wedge p' = \frac{2}{h}\,q\wedge q' - q\wedge p = (2-Y)M_h$ and $q'\wedge p = \frac{2}{h}\,q\wedge q' - q'\wedge p' = (2-Y')M_h$. Conservation of c_h gives $|q|^2 = \tilde{y} + \frac{h^2}{2}K - \frac{h^2}{4}u_2\tilde{y}^2 - g$ and the same expression with primes, with the same constant g. Substitute these into (11); the terms in KK' cancel. With $\lambda = \frac{1}{2}\kappa|M_h|^2$, $s = 2-Y = 1-\frac{1}{2}\kappa\tilde{y}$ and $\mu = 2-\kappa g$, what remains is $K'(\mu - s^2) - \lambda s^2 = K(\mu - s'^2) - \lambda s'^2$, that is,

$$(K'+\lambda)(\mu - s^2) = (K+\lambda)(\mu - s'^2).$$

Finally $\mu - s^2 = 1 + \kappa\tilde{y} - \frac{1}{4}\kappa^2\tilde{y}^2 - \kappa g = D$ and $\mu - s'^2 = D'$, so $E'_h = E_h$.

(c) Put $\xi = q - \frac{h}{2}p$, $\eta = q + \frac{h}{2}p$ and $\eta' = q' + \frac{h}{2}p'$. The first equation in (10) says $\eta = q' - \frac{h}{2}p'$, so $q = \frac{1}{2}(\xi+\eta)$ and $q' = \frac{1}{2}(\eta+\eta')$. Let $A = \xi\cdot\eta$, $B = \eta\cdot\eta'$ and $C = |\eta|^2$. In these variables the third equation in (10) reads $y' - y = B - A$, and the second one reads $Y\eta' + Y'\xi = (4 - Y - Y')\,\eta$. Hence

$$Y' - Y = \tfrac{1}{2}\kappa(B-A), \qquad YB + Y'A = (4-Y-Y')\,C. \tag{12}$$

Moreover $D = (2-Y)Y + \kappa A$ and $D' = (2-Y')Y' + \kappa B$, because $1 - \frac{1}{4}\kappa^2\tilde{y}^2 = (2-Y)Y$ and $|q|^2 - \frac{h^2}{4}|p|^2 = \xi\cdot\eta$. A Schur-complement computation in the block form of $\mathrm{Id} \mp \frac{h}{2}F'$ gives

$$\begin{aligned}\det\big(\mathrm{Id} - \tfrac{h}{2}F'(z)\big) &= Y^{n-1}\big(Y + \kappa\, q\cdot\eta\big) = Y^{n-1}\big(Y + \tfrac{1}{2}\kappa(A+C)\big),\\ \det\big(\mathrm{Id} + \tfrac{h}{2}F'(z')\big) &= Y'^{n-1}\big(Y' + \kappa\, q'\cdot\eta\big) = Y'^{n-1}\big(Y' + \tfrac{1}{2}\kappa(B+C)\big).\end{aligned}$$

Using (12), a short computation shows

$$2D = (4-Y-Y')\big(Y + \tfrac{1}{2}\kappa(A+C)\big), \qquad 2D' = (4-Y-Y')\big(Y' + \tfrac{1}{2}\kappa(B+C)\big).$$

For the first identity, expand the right-hand side, replace $(4-Y-Y')C$ by $YB + Y'A$, and then $\frac{1}{2}\kappa(B-A)$ by $Y'-Y$; the second follows by symmetry. Where $4-Y-Y' \neq 0$, dividing the two identities and using Proposition 2.1(d) gives $\det D\Phi_h = (Y'/Y)^{n-1}D'/D$, which is equivalent to $P_n(z') = \det D\Phi_h\, P_n(z)$. Where $4-Y-Y' = 0$, both D and D' vanish and the measure identity holds trivially. □

The integrals and the measure were first found by exact searches for Darboux polynomials (Appendix B). The proofs explain their form. The angular momenta come from the block theorem with non-commuting fields. The energy comes from a block $(p,\tilde{y})$ that fails the closure condition only by the term $|q\wedge p|^2$, which the conserved discrete angular momentum absorbs. The measure density is Y^{n-1} times the denominator of E_h. For $M = 0$, E_h is a function of the two-variable integral $K/(1+\kappa|q|^2)$ of Corollary 3.3. All identities were also verified independently by computer algebra (Appendix A).

Corollary 6.2 (integrability)**.** *For $n = 2$ the integrals c_h, M_h, E_h are functionally independent. Let L be a compact connected component of a regular level set with $M_h \neq 0$ on which $\Phi_h^{\pm1}$ are defined and $P_2 \neq 0$, and assume $\Phi_h(L) = L$ and that the induced circle map below preserves orientation (both hold for small h). Then L is a 2-torus and $\Phi_h|_L$ is smoothly conjugate to a skew rotation*

$$(\theta,\phi) \mapsto (\theta + \rho,\ \phi + g(\theta)), \qquad (\theta,\phi) \in \mathbb{T}^2.$$

If ρ is Diophantine, $\Phi_h|_L$ is conjugate to a translation of $\mathbb{T}^2$, i.e. the motion is quasi-periodic. For $n > 2$ the same holds on the level sets of c_h, E_h and all $M_{ab,h}$, which comprise $2n-1$ independent integrals on $\mathbb{R}^{2n+1}$.

Proof. A 3×3 minor of the Jacobian of (c_h, M_h, E_h) is a non-zero rational function (Appendix A), which gives functional independence. L is a compact, orientable 2-manifold, being a regular level set in $\mathbb{R}^5$. Rotations of the pair (q,p) preserve the integrals, so the connected group $SO(2)$ maps L to itself; it acts freely because $M \neq 0$. A compact orientable surface with a free circle action is a torus, and $L/SO(2)$ is a circle. Φ_h commutes with the action, so it induces a homeomorphism f of $L/SO(2)$. The measure $\mathrm{d}z/P_2$ induces on L the invariant density $\sigma = (\mathrm{d}z/P_2)/(\mathrm{d}c_h \wedge \mathrm{d}M_h \wedge \mathrm{d}E_h)$, which is smooth, nowhere zero and rotation invariant, so $|\sigma|$ descends to a smooth positive f-invariant density on the circle. Its distribution function conjugates f to a rigid rotation by ρ. Since $M \neq 0$, $q \neq 0$ on L, so $\phi = \arg q$ (in the plane of motion) is a coordinate on which $SO(2)$ acts by translation; together with a coordinate θ on the circle $L/SO(2)$ in which f is the rotation, it gives coordinates (θ,ϕ) on L. Since Φ_h commutes with the translations of ϕ, it has the stated form, with $g(\theta) = \arg q' - \arg q$. By (a), $q \wedge q' = hM_h \neq 0$, so this increment stays in $(0,\pi)$ or in $(-\pi,0)$, and g is a smooth real-valued function. For Diophantine ρ the cohomological equation $\gamma(\theta+\rho) - \gamma(\theta) = g(\theta) - \bar{g}$ has a smooth solution, and $\phi \mapsto \phi - \gamma(\theta)$ conjugates to the translation by $(\rho, \bar{g})$. For $n > 2$, fixing all $M_{ab,h}$ fixes the oriented plane Π (the bivector M_h is decomposable, giving $2n-3$ independent components) and reduces to $n = 2$. □

Proposition 6.3 (bounded orbits)**.** *Let $u_2 > 0$. If $D(z_0) > 0$, for instance if $h^4 u_2 K(z_0) < 2$, then the orbit of z_0 under Φ_h is bounded, as long as it is defined.*

Proof. Let $e = E_h(z_0) \geq 0$ and $c = c_h(z_0)$. Along the orbit $K + \frac{1}{2}\kappa|M_h|^2 = eD$, hence $K\big(1 + \frac{1}{2}eh^4u_2\big) \leq e\big(1 + \kappa|q|^2\big)$, that is, $K \leq e_*\big(1 + \kappa|q|^2\big)$ with $e_* = e/(1 + \frac{1}{2}eh^4u_2)$ and $e_* h^4 u_2 < 2$. Put $s = \frac{h^2}{4}|p|^2 \geq 0$ and $t = |q|^2 - s$, so that $y = c + t$. The inequality becomes

$$s\Big(\frac{2}{h^2} - e_* h^2 u_2\Big) + \frac{u_2}{2}\Big(t + c + \frac{u_1}{u_2}\Big)^2 - e_* h^2 u_2\, t \leq e_*.$$

The coefficient of s is positive, so the inequality bounds t (a quadratic with positive leading coefficient) and then s. Hence $|q|^2 = t + s$, $|p|^2$ and y stay bounded. □

Remark 6.4 (an $O(n)$ explicit form). Eliminating q', p' from Kahan's equations leaves one scalar linear equation for y'. With $\alpha = hU'(y)$, $Y = 1 + \frac{h}{2}\alpha$ and $\eta = q + \frac{h}{2}p$,

$$y' = \frac{yY + h\big(2p - (\alpha + hu_1)q\big)\cdot\eta}{Y + \kappa\, q\cdot\eta},$$
$$q' = \frac{(1 - \frac{h}{2}\alpha')q + hp}{Y}, \qquad p' = \frac{(1 - \frac{h}{2}\alpha)p - (\alpha + \alpha')q}{Y},$$

with $\alpha' = hU'(y')$. A step therefore costs $O(n)$ operations, like one Störmer–Verlet step, and no linear system has to be solved. The denominators vanish exactly where $\det(\mathrm{Id} - \frac{h}{2}F') = Y^{n-1}(Y + \kappa\, q\cdot\eta)$ does. Two further identities make the link with discrete Newtonian equations [30]. Along an orbit (q_k, p_k, y_k),

$$q_k \wedge q_{k+1} = h\, M_h, \qquad q_{k+1} + q_{k-1} = \lambda_k\, q_k, \quad \lambda_k = \frac{2 - \frac{1}{2}h^2\big(U'(y_{k+1}) + U'(y_{k-1})\big)}{1 + \frac{1}{2}h^2 U'(y_k)}.$$

The first identity exhibits M_h as the discrete angular momentum, and the second is a "central force" recurrence in which the force is evaluated through the auxiliary sequence y_k. With the initial value $y_0 = |q_0|^2 - \frac{h^2}{4}|p_0|^2$ on the shifted sheet, (q_k, p_k) is a symmetric, second-order, integrable discretisation of the original system on $\mathbb{R}^{2n}$ (Theorem 4.4).

6.1 Radial dynamics: a QRT map and an explicit solution

By Remark 6.4, $q_k \cdot q_{k+1} = \big((1-\frac{h}{2}\alpha_{k+1})|q_k|^2 + h\, q_k \cdot p_k\big)/Y_k$ and $|q_k \wedge q_{k+1}| = hm$ with $m = |M_h|$. So in the plane of motion the polar angle of q_k advances by ϑ_k with

$$\tan\vartheta_k = \frac{hm\, Y_k}{(1-\frac{h}{2}\alpha_{k+1})|q_k|^2 + h\, q_k \cdot p_k}, \tag{13}$$

and the whole orbit is determined by the radial quantities y_k, $|q_k|^2$ and $b_k = q_k \cdot p_k$. The next theorem shows that they are elliptic functions of k, generated by a QRT map [16, 17, 18].

Theorem 6.5 (radial dynamics)**.** *Let $u_1 = 0$ (otherwise replace y by $\tilde y = y + u_1/u_2$ and c_h by $c_h + u_1/u_2$), and let (z_k) be an orbit with $c_h = \gamma$, $|M_h|^2 = m^2$ and $E_h = e$. Define the symmetric biquadratic polynomial*

$$\begin{aligned}\mathcal{Q}(x,x') = P\sigma_1^2 - 4\kappa\big(e\kappa h^2+2\big)\sigma_1\sigma_2 + 16\big(e\kappa h^2+1\big)\sigma_2 \\ + 16eh^2\sigma_1 - 16h^2\big(2e\gamma + m^2\big),\end{aligned}$$

where $\sigma_1 = x + x'$, $\sigma_2 = xx'$ and $P = 4(\kappa\gamma - 1) + \kappa h^2(2e\gamma\kappa - 4e + \kappa m^2)$.

(a) $\mathcal{Q}(y_k, y_{k+1}) = 0$ for all k. For fixed γ and m, $\mathcal{Q}$ depends linearly on e. Hence y_{k+1} and y_{k-1} are the two roots of $\mathcal{Q}(y_k,\cdot)$, and $(y_{k-1}, y_k) \mapsto (y_k, y_{k+1})$ is the symmetric QRT map of the pencil of biquadratic curves $\{\mathcal{Q} = 0\}$ parametrised by e.

(b) $|q_k|^2 = \mathcal{A}(y_k)$ and $b_k^2 = Q_4(y_k)$, where

$$\begin{aligned}\mathcal{A}(x) &= \big(1 + \tfrac{1}{2}e\kappa h^2\big)\big(x - \gamma - \tfrac{1}{4}\kappa x^2\big) + \tfrac{1}{2}eh^2 - \tfrac{1}{4}\kappa h^2 m^2,\\ Q_4(x) &= \tfrac{4}{h^2}\mathcal{A}(x)\big(\gamma - x + \mathcal{A}(x)\big) - m^2\big(1 + \tfrac{1}{2}\kappa x\big)^2.\end{aligned}$$

If Q_4 has simple roots and h is small enough, there are x_0 and τ such that $(y_k, b_k) = \big(\mathfrak{y}(x_0 + k\tau), \mathfrak{y}'(x_0 + k\tau)\big)$ for all k, where $\mathfrak{y}$ is the elliptic function with $(\mathfrak{y}')^2 = Q_4(\mathfrak{y})$.

(c) As $h \to 0$, $\mathcal{Q}(x, x+hw)/h^2 \to -4w^2 + 16\big[(x-\gamma)(2e - u_2x^2) - m^2\big]$, so the curve $\mathcal{Q} = 0$ is a deformation of the radial energy relation $\dot y^2 = 4\big[(y-\gamma)(2e - u_2y^2) - m^2\big]$ of the continuous system.

Proof. (a) Use ξ, η, η' from the proof of Theorem 6.1(c). There we saw $|q|^2 - \frac{h^2}{4}|p|^2 = \xi \cdot \eta$, so $\xi \cdot \eta = y - \gamma$, and likewise $\eta \cdot \eta' = y' - \gamma$ because $\xi' = \eta$. The second relation in (12) gives $|\eta|^2 = \big(Y(y'-\gamma) + Y'(y-\gamma)\big)/(4 - Y - Y')$. Since $q \wedge p = \xi \wedge \eta/h$, conservation of M_h and Lagrange's identity give $|\xi|^2|\eta|^2 = (y-\gamma)^2 + h^2Y^2m^2$. With $q = \frac{1}{2}(\xi + \eta)$ and $p = (\eta - \xi)/h$, the quantities $|q|^2$ and $|p|^2$, and hence $E_h(z)$, become rational functions of y and y'. A direct computation (Appendix A) shows that the numerator of $E_h(z) - e$ is a constant multiple of $D(z)\,\mathcal{Q}(y, y')$. Since $D(z) \neq 0$, $\mathcal{Q}(y, y') = 0$. Symmetry and linearity in e are visible in the formula, and the QRT statement follows because both $\mathcal{Q}(y_{k-1}, y_k)$ and $\mathcal{Q}(y_k, y_{k+1})$ vanish.

(b) On the level set, $c_h = \gamma$ gives $|p|^2 = \frac{4}{h^2}(\gamma - y + |q|^2)$, and then $E_h = e$ is linear in $|q|^2$, with root $\mathcal{A}(y)$. Lagrange's identity gives $b^2 = |q|^2|p|^2 - |q \wedge p|^2 = Q_4(y)$. So the map induced on the orbit space is a birational automorphism of the curve $\mathfrak{E} = \{(y, b) : b^2 = Q_4(y)\}$, which has genus one when Q_4 has simple roots. In a uniformising coordinate x with $\mathrm{d}x = \mathrm{d}y/b$, every automorphism of $\mathfrak{E}$ is $x \mapsto \zeta x + \tau$ with ζ a root of unity. The induced map depends continuously on h and tends to the identity as $h \to 0$, so $\zeta = 1$: it is a translation. Then $\mathfrak{y}(x) = y$ satisfies $\mathfrak{y}' = b$ and $(\mathfrak{y}')^2 = Q_4(\mathfrak{y})$.

(c) Expand $\mathcal{Q}(x, x + hw)$ in powers of h (Appendix A). □

Table 1: Degree growth $d_1, d_2, \ldots$ of lifted Kahan maps (exact, modulo a prime, along a random line). Polynomial (here linear or quadratic) growth indicates integrability, exponential growth chaos. "Source" names the origin of the continuous system's integrability. Lifts use $y = |q|^2$ for isotropic systems, $y_k = q_k^2$ (or the products noted) for quartic potentials, and exponential or harmonic variables otherwise.

Lifted system	$d_1, d_2, \ldots$	Source of integrability
Integrable lifted maps		
quartic oscillator ($\mathbb{R}^3$)	$3, 9, 19, 33, 51, 73, 99$	one degree of freedom
isotropic quartic, $n = 2, 3, 4$	$4, 12, 25, 44, 68, 97, 132$	rotations, one d.o.f.
isotropic quartic plus harmonic term	$4, 12, 25, 44, 68, 97$	rotations, one d.o.f.
isotropic quartic, magnetic field, $n = 2$	$4, 13, 28, 49, 76, 109$	rotations, one d.o.f.
decoupled $q_1^4 + q_2^4$	$5, 17, 37, 65, 101, 145$	decoupling
1:6:1 quartic, lift in $q_1 \pm q_2$	$5, 17, 37, 65, 101, 145$	decoupling after rotation
two-particle Toda	$2, 4, 6, 8, 10, 12$	translation, one d.o.f.
two coupled rotors	$3, 9, 19, 33, 51, 73$	translation, one d.o.f.
Non-integrable lifted maps		
1:6:1 quartic, lift $y_k = q_k^2$	$5, 25, 113, > 400$	same system, unadapted lift
quartic 1:6:8	$5, 25, 113, > 400$	hidden integral, quartic in momenta
quartic 1:12:16	$5, 25, 113, > 400$	parabolic separation
Garnier, $a_1 \neq a_2$	$4, 16, 58, 208, > 400$	elliptic separation
open Toda, three particles	$3, 9, 23, 57, 139, 337, 815$	Lax pair
anisotropic $(q_1^2 + 2q_2^2)^2$	$4, 16, 58, 208, 742$	none (not integrable)
sextic isotropic	$5, 16, 43, 114, 305, > 400$	lift not fully protected
non-symmetric quartic, one d.o.f.	$3, 9, 23, 57, 139, 337, 815$	energy not protected
prolonged Euler top ($v = \dot{x}$)	$4, 16, 52, 160, > 400$	integrals not protected

Corollary 6.6 (explicit solution)**.** *Under the hypotheses of Theorem 6.5(b), in the plane of motion*

$$q_k = \sqrt{\mathcal{A}\big(\mathfrak{y}(x_0 + k\tau)\big)}\,\big(\cos\theta_k, \sin\theta_k\big), \qquad \theta_{k+1} = \theta_k + \vartheta_k,$$

with ϑ_k given by (13) in terms of $y_k = \mathfrak{y}(x_0 + k\tau)$, y_{k+1} and $b_k = \mathfrak{y}'(x_0 + k\tau)$. The radial motion of the discrete system is thus the continuous elliptic motion of the modified curve $\mathfrak{E}$, sampled at equal steps τ of its uniformising parameter; for small h, $\tau \approx 2h$.

Kahan discretisations of planar quadratic systems are known to produce QRT-type maps and translations on elliptic curves [13, 15]. Here the QRT map arises as the radial part of a map in five or more dimensions, after reduction by the rotation group. Numerically, the points (y_k, y_{k+1}) lie on $\mathcal{Q} = 0$ to round-off (Fig. 2(c)). Along an orbit with $n = 3$ and $h = 0.4$, the pairs (y_k, b_k) agree with $(\mathfrak{y}(x_0 + k\tau), \mathfrak{y}'(x_0 + k\tau))$, $\tau = 0.7822$, to 4×10^{-9} over 60 steps, which is the accuracy of the reference integration of $\mathfrak{y}'' = \frac{1}{2}Q_4'(\mathfrak{y})$.

7 Which integrable systems survive lifting?

Proposition 4.6 and Theorem 6.1 produce integrable maps from systems whose integrability has a simple origin: one degree of freedom, possibly after reduction by a linear symmetry. To see how far this goes, we computed the degree growth of iterates for nineteen lifted Kahan maps. Kahan maps are birational; restricted to a generic line, the k-th iterate has some degree d_k, and zero algebraic entropy $\lim_{k\to\infty} \frac{1}{k} \log d_k = 0$, in practice polynomial growth of d_k, is the standard indicator of integrability [34, 35]. We computed d_k exactly, modulo a large prime (Appendix B). Table 1 and Fig. 1 summarise the results. Every map in the upper half of Table 1 is proved to be integrable in Section 7.1, so for these maps the degree growth is a consistency check. For the maps in the lower half, exponential growth is strong evidence that they are not integrable.

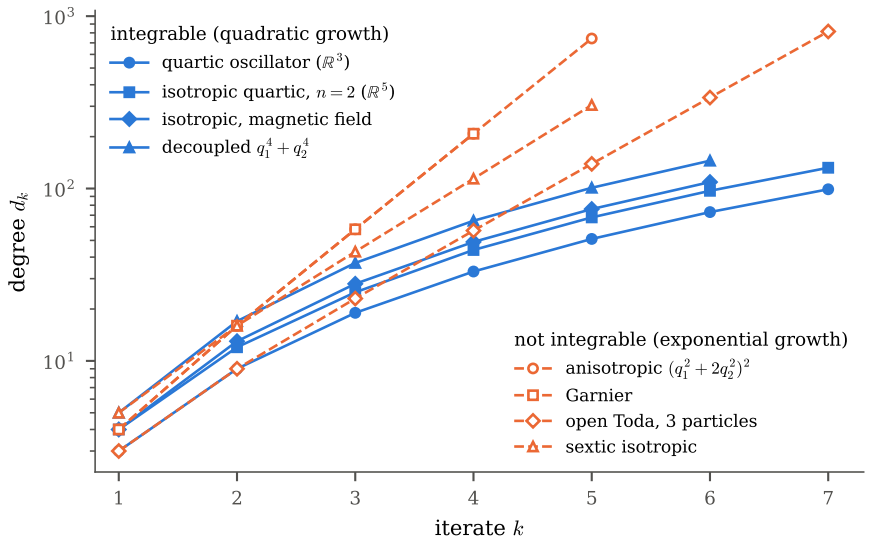


Figure 1: Degree growth of representative lifted Kahan maps (logarithmic scale). Integrable maps (blue, solid) grow quadratically; non-integrable ones (orange, dashed) grow exponentially. The Garnier sequence coincides with the anisotropic one for $k \le 4$.

7.1 Proved cases

Two elementary facts extend the results of the previous sections. Kahan's method commutes with affine changes of variables (Proposition 2.1(c)), and it acts factor by factor on products: if $F(z_1, z_2) = \big(F_1(z_1), F_2(z_2)\big)$, then the equations (1) decouple, and Kahan's map of F is the product of the Kahan maps of F_1 and F_2.

Proposition 7.1 (proved integrable cases)**.** *For each of the following lifted systems, Kahan's map preserves a measure with an explicit density and conserves explicit, functionally independent modified integrals. Their common level sets are curves in cases (a) and (d), surfaces that carry a circle action commuting with the map in cases (b) and (e), and products of these in case (c).*

(a) One degree of freedom: the systems of Proposition 4.6, which include all even quartic oscillators.

(b) The isotropic family of Theorem 6.1, in every dimension, and every lifted system that is affinely conjugate to it. Examples are $H = \frac{1}{2}p^\top G p + U(q^\top G^{-1} q)$ with G symmetric positive definite, lifted with $y = q^\top G^{-1} q$, and the two-body problem $H = \frac{1}{2}|p_1|^2 + \frac{1}{2}|p_2|^2 + U(|q_1 - q_2|^2)$ with $q_k, p_k \in \mathbb{R}^n$, lifted with $y = |q_1 - q_2|^2$.

(c) Lifted systems that are affinely conjugate to a product of systems from the other cases. Examples are the potential $q_1^4 + q_2^4$ with $y_k = q_k^2$, and $q_1^4 + 6q_1^2 q_2^2 + q_2^4$ with $y_\pm = (q_1 \pm q_2)^2$.

(d) A single exponential or a single harmonic, $H = \frac{1}{2}p^\top G p + c\,\mathrm{e}^{\alpha \cdot q}$ or $H = \frac{1}{2}p^\top G p + g\cos(\beta \cdot q)$ on $\mathbb{R}^n \times \mathbb{R}^n$, with G symmetric and the lift of Theorem 5.1. Examples are the two-particle Toda chain and two coupled rotors.

(e) The planar isotropic oscillator in a uniform magnetic field (Proposition 7.2 below).

Proof. (a) is Proposition 4.6: two integrals on $\mathbb{R}^3$. (b) follows from Theorem 6.1 and Proposition 2.1(c). For the first example, the linear change $\tilde{q} = G^{-1/2} q$, $\tilde{p} = G^{1/2} p$ gives the isotropic system with $y = |\tilde{q}|^2$. For the two-body problem, put $r = q_1 - q_2$, $\pi = p_1 - p_2$, $R = q_1 + q_2$ and $P = p_1 + p_2$. The lifted field becomes $\dot{r} = \pi$, $\dot{\pi} = -4U'(y)\, r$, $\dot{y} = 2\, r \cdot \pi$, which is (9) with U replaced by $2U$, together with $\dot{R} = P$, $\dot{P} = 0$, on which Kahan's map is the exact flow. (c)

The map is the product of the maps of the factors. In the second example the lifted field is $\dot{p}_1 = -2(q_1+q_2)y_+ - 2(q_1-q_2)y_-$, $\dot{p}_2 = -2(q_1+q_2)y_+ + 2(q_1-q_2)y_-$, $\dot{y}_\pm = 2(q_1 \pm q_2)(p_1 \pm p_2)$, which decouples in the variables $q_1 \pm q_2$, $p_1 \pm p_2$, $y_\pm$ into two quartic oscillators. (d) On $\mathbb{R}^N$ with $N = n+1$ (exponential) or $N = n+2$ (harmonic), Theorem 5.1 gives the Liouville measure, E_h and, in the harmonic case, the modified circle $\rho_h = (C^2+S^2)/(1+\frac{1}{4}h^2b^2)$ with $b = \beta^\top G p$. Since $\dot{p}$ is a multiple of α (resp. β), the $n-1$ linear functions $w\cdot p$ with $w \perp \alpha$ (resp. $w \perp \beta$) are integrals of the lifted field, hence of the map (Proposition 2.1(e)). These $N-1$ integrals are functionally independent: the linear ones do not involve the lifted variables, and $\partial E_h/\partial u = c - \frac{1}{4}h^2c^2u\,\alpha^\top G\alpha$ (exponential), resp. $\det \partial(E_h,\rho_h)/\partial(C,S) = 2gS\big(1+\frac{1}{4}h^2gC\,\beta^\top G\beta\big)/\big(1+\frac{1}{4}h^2b^2\big)$ (harmonic), is not identically zero. (e) is Proposition 7.2. □

In each case the map is integrable in the usual sense. On a compact regular level set that is a curve, the preserved measure induces an invariant density, and the argument in the proof of Corollary 6.2 shows that the map is conjugate to a rotation of the circle. In cases (b) and (e) the same argument applies to the circle $L/SO(2)$ and gives a skew rotation of a 2-torus, under the hypotheses of Corollary 6.2. Level sets need not be compact: for the two-particle Toda chain they are open arcs, traversed by scattering orbits.

Proposition 7.2 (the magnetic case)**.** *Let $n = 2$ and $U(y) = \frac{1}{2}u_2y^2$. A charged particle in the potential $U(|q|^2)$ and a uniform magnetic field of strength B, lifted with $y = |q|^2$, gives the quadratic field*

$$\dot{q} = v, \qquad \dot{v} = -2U'(y)\,q + B\,\mathbb{J}v, \qquad \dot{y} = 2\,q\cdot v, \qquad \mathbb{J} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{14}$$

Write $\beta = \frac{1}{2}hB$, $\kappa = h^2u_2$, $\alpha = hU'(y)$, $Y = 1+\frac{1}{2}\kappa y$, $a = |q|^2$, $M = q\wedge v$ and $K = \frac{1}{2}|v|^2 + \frac{1}{2}u_2y^2$, and let $c_h = y - |q|^2 + \frac{h^2}{4}|v|^2$ and $D = (2-Y)Y + \kappa(y-c_h)$ be as in Theorem 6.1.

(a) In the complex notation $Q = q_1 + \mathrm{i}q_2$, $V = v_1 + \mathrm{i}v_2$, Kahan's map is

$$Q' = \frac{(2-Y'-\mathrm{i}\beta)\,Q + hV}{Y - \mathrm{i}\beta}, \qquad V' = \frac{-(\alpha+\alpha')\,Q + (2-Y+\mathrm{i}\beta)\,V}{Y-\mathrm{i}\beta},$$

where y' solves the scalar linear equation $y' - y = h\,(q\cdot v' + q'\cdot v)$. It satisfies the discrete Larmor identity

$$Y\,q\wedge q' - \beta\,q\cdot q' = h\big(q\wedge v - \tfrac{1}{2}B|q|^2\big). \tag{15}$$

(b) It conserves c_h,

$$L_h = \frac{P_L}{P^B} \qquad \text{and} \qquad \mathcal{E}_h = \frac{\hat{P}_E}{(2+\kappa c_h)\,P^B},$$

and preserves the measure $\mathrm{d}z/P^B$, where

$$\begin{aligned} P^B &= Y(D-\beta^2) + 2\beta^2 + h\beta\kappa M, \\ P_L &= M(D+\beta^2) - B\big[2aY + (1+Y)\,c_h + \tfrac{1}{2}y\,(Y^2 - 2Y - 2 + \beta^2)\big], \\ \hat{P}_E &= (2+\kappa c_h)\,KY + \tfrac{1}{2}\kappa M^2(4-Y) + \tfrac{1}{2}B^2\big(y - (a+c_h)(2-Y)\big) - \kappa BM\big(\tfrac{1}{2}y + a + c_h\big). \end{aligned}$$

The integrals c_h, L_h and $\mathcal{E}_h$ are functionally independent.

(c) For $B = 0$, $P^B = YD = P_2$, $L_h = M_h$ and $\mathcal{E}_h = E_h + \kappa M_h^2/\big(2(2+\kappa c_h)\big)$. As $h \to 0$, $L_h \to q\wedge v - \frac{1}{2}By$ and $\mathcal{E}_h \to K$, the canonical angular momentum (in the symmetric gauge) and the energy.

Proof. (a) By Proposition 2.1(b), Kahan's equations are $q'-q=\frac{h}{2}(v+v')$, $v'-v=-\alpha q'-\alpha' q+\beta\mathbb{J}(v+v')$ and $y'-y=h(q\cdot v'+q'\cdot v)$. Since $\mathbb{J}$ acts as multiplication by i, the first two read $\left(\begin{smallmatrix}1 & -h/2\\ \alpha & 1-\mathrm{i}\beta\end{smallmatrix}\right)\binom{Q'}{V'}=\left(\begin{smallmatrix}1 & h/2\\ -\alpha' & 1+\mathrm{i}\beta\end{smallmatrix}\right)\binom{Q}{V}$. The matrix on the left has determinant $Y-\mathrm{i}\beta$, and inverting it gives the formulas. Multiply the formula for Q' by $(Y-\mathrm{i}\beta)\bar{Q}$ and take imaginary parts: since $\bar{Q}Q'=q\cdot q'+\mathrm{i}\,q\wedge q'$ and $\mathrm{Im}(\bar{Q}V)=q\wedge v$, this gives (15).

(b), (c) The map is equivariant and all functions involved are invariant under rotations of the plane, so it suffices to verify the identities on the slice $q_2=0$. There they are identities between rational functions of (q_1,v_1,v_2,y) and (h,B,u_2), which we verified by computer algebra (Appendix A). The determinants in the measure identity are $\det\big(\mathrm{Id}\mp\frac{h}{2}F'(z)\big)=Y\big(Y+\kappa\,q\cdot(q\pm\frac{h}{2}v)\big)+\beta^2+\frac{1}{2}h\beta\kappa M$. Independence follows from a non-zero 3×3 minor of the Jacobian matrix. □

The integrals were found by the exact search of Appendix B, run in the basis of rotation-invariant monomials $y^i a^j b^k c^l M^m$ with $b=q\cdot v$ and $c=|v|^2$, modulo several primes and with rational reconstruction of the coefficients. They explain the quadratic degree growth in Table 1. Proposition 7.2 also shows that Theorem 3.2 is sufficient but not necessary for protection: in the velocity variables the canonical angular momentum fails the closure condition of Theorem 3.2, yet it has the modified integral L_h. The Larmor identity (15) plays the role of $q\wedge q'=hM_h$ in Remark 6.4. Applied to the inverse step Φ_{-h}, it gives $Y'\,q\wedge q'-\beta\,q\cdot q'=h\big(q'\wedge v'-\frac{1}{2}B|q'|^2\big)$, so the continuous angular momentum changes by $\frac{\kappa}{2h}(y'-y)\,q\wedge q'$ per step. In the continuous system, Larmor's theorem removes the magnetic field of a central-force problem by passing to a frame that rotates with angular velocity $\frac{1}{2}B$, at the price of an extra harmonic term $\frac{1}{8}B^2|q|^2$. We have not found a discrete analogue of this theorem that would derive Proposition 7.2 from Theorem 6.1.

7.2 Hidden symmetries and a conjecture

Three observations stand out. First, every integrable map in Table 1 is covered by Proposition 7.1; in particular the isotropic family stays integrable with a harmonic term, in every dimension, and in a uniform magnetic field. Second, in our tests every lifted system whose continuous integrability rests on a hidden symmetry is chaotic after lifting: three-particle Toda, the Garnier system [37], and the quartic potentials 1:6:8 and 1:12:16 [36]. Third, the separable 1:6:1 potential $q_1^4+6q_1^2q_2^2+q_2^4=\frac{1}{2}[(q_1+q_2)^4+(q_1-q_2)^4]$ gives an integrable map with a lift in the separating coordinates $q_1\pm q_2$ and a chaotic map with the lift $y_k=q_k^2$. Kahan's method commutes with linear changes of variables, but these two lifts are not related by one, so the lift must respect the linear structure that makes the system integrable.

Conjecture 7.3. *Let $\dot{x}=f(x)$ be integrable because of a group of linear symmetries together with a reduction to one degree of freedom, and let F be a quadratic lift that is equivariant under the symmetries, adapted to the reduction and has only protected constraints. If the reduced one-degree-of-freedom problem lies in a class on which Kahan's method is integrable (Proposition 4.6, or Proposition 7.1(d)), then the lifted Kahan map is integrable.*

Proposition 7.1 confirms the conjecture in every case of this kind that we tested. In cases (b) and (e) the symmetry group consists of rotations and the reduction is to the radial motion; in cases (c) and (d) the reduction is an affine change of variables that splits off decoupled or free factors. What is missing for a proof is a reduction theory for Kahan maps with linear symmetries (Section 9).

The heuristic reason is that every mechanism producing exact modified integrals here is linear-algebraic: Kahan's method commutes with linear maps, and the modified integrals are built from quadratic invariants of linear flows and their speeds. A linear symmetry therefore passes to the map, while hidden integrals (Toda's $\mathrm{tr}\,L^3$, Garnier's Uhlenbeck integrals, the quartic

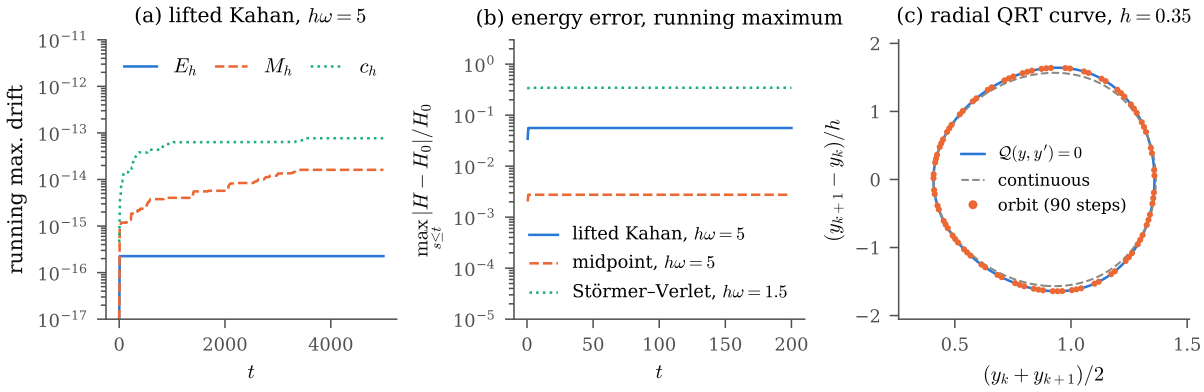


Figure 2: (a), (b) Stiff isotropic oscillator in $\mathbb{R}^3$, $\omega = 10$. (a) Lifted Kahan at $h\omega = 5$: running maximum of the drift of the modified invariants over 10^4 steps (relative for E_h and M_h, absolute for c_h). (b) Running maximum of the relative energy error. (c) Radial dynamics of an orbit with $n = 2$, $u_2 = 1$, $u_1 = 0$, $h = 0.35$, $q_0 = (1.2, 0)$, $p_0 = (0.3, 0.9)$: the points $\left(\frac{1}{2}(y_k + y_{k+1}), (y_{k+1} - y_k)/h\right)$ lie on the curve $\mathcal{Q}(y, y') = 0$ of Theorem 6.5 (solid), a deformation of the continuous radial relation $\dot{y}^2 = 4[(y - \gamma)(2e - u_2 y^2) - m^2]$ with the same γ, e, m (dashed).

integrals of 1:6:8) are not invariants of linear flows and nothing protects them. This is a statement about *lifted* systems only: the genuinely quadratic Clebsch and $\mathfrak{so}(4)$ Manakov tops keep hidden quadratic integrals under Kahan's method [5, 6].

8 Numerical experiments

We test the isotropic family as an integrator for a stiff anharmonic oscillator,

$$H = \tfrac{1}{2}|p|^2 + \tfrac{1}{2}\omega^2|q|^2 + \tfrac{1}{4}|q|^4, \qquad q, p \in \mathbb{R}^3, \quad \omega = 10,$$

i.e. $u_2 = \frac{1}{2}$ and $u_1 = \frac{1}{2}\omega^2$, with $q_0 = (0.8, 0.1, -0.3)$, $p_0 = (0, 3, 1)$ and y_0 on the shifted sheet. The stiff harmonic term sits in the linear part of the lift, which Kahan's method treats like the midpoint rule. We compare with three methods: the implicit midpoint rule on (q, p), solved by Newton's method to a tolerance of 10^{-14}; the Störmer–Verlet method; and Strang splitting into the exact harmonic flow and a kick by the quartic force (the impulse method), which is the natural explicit method for this problem when the stiff part is linear and known [44, 45, 46]. Figures 2 and 3 show the results.

- *Exact invariants.* At $h\omega = 5$ the modified invariants E_h, M_h and c_h drift by at most 2.2×10^{-16}, 1.6×10^{-14} and 7.7×10^{-14} over 10^4 steps, i.e. by round-off. The true energy error stays bounded at 5.6×10^{-2}.

- *Resonances.* Störmer–Verlet is unstable for $h\omega > 2$. The impulse method is stable for most step sizes, but it resonates near the integer multiples of $\pi/2$, as is well known for impulse and multiple-time-stepping methods [44, 45]: of the 1961 step sizes, 100 gave errors above 0.3, 50 of them escaping orbits, all in six narrow bands at $h\omega \approx k\pi/2$, $k = 1, \ldots, 6$. The lifted Kahan map has a single failure band, $h\omega \in [1.995, 2.005]$ (three step sizes, two of them escaping). Its location is predicted by the theory: $D(z_0)$ changes sign at $h\omega = 1.99998$, and Proposition 6.3 guarantees bounded orbits below that value. At $h\omega \approx 2$ the Cayley transform of the harmonic part, which Kahan's method shares with the midpoint rule, rotates by $\pi/2$ per step (here $Y \approx 1 + \frac{1}{4}h^2\omega^2 = 2$), a 1:4 resonance, and the midpoint rule also shows a bump there. Beyond the band, $D(z_0) < 0$ and Proposition 6.3 no longer applies, but all orbits in the scan stayed bounded. Larger amplitudes can escape there: one of 20 random orbits with $|q_0| = 3$ and $|p_0| = \omega|q_0|$ escaped at $h\omega = 10$.

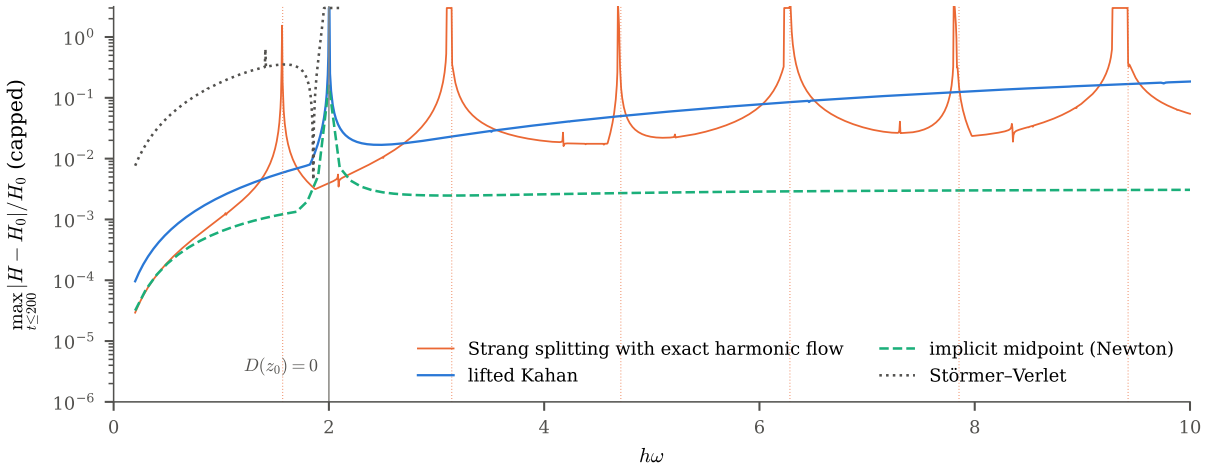


Figure 3: Maximal relative energy error on $[0, 200]$ against $h\omega$ for the stiff oscillator, on a grid of 1961 step sizes (99 for the midpoint rule); errors are capped at 3, and escaping orbits are drawn at the cap. Dotted vertical lines: integer multiples of $\pi/2$, where the impulse method resonates. Solid vertical line: the step size at which $D(z_0) = 0$; Proposition 6.3 guarantees bounded lifted Kahan orbits to its left.

- *Accuracy and cost.* For $h\omega \le 1$ the lifted Kahan map is about 75 times more accurate than Störmer–Verlet (median error ratio 0.013), because, like the midpoint rule, it conserves the energy of the harmonic part exactly, and about three times less accurate than the midpoint rule. For $h\omega \in [2.2, 10]$ the median errors are 0.081 (lifted Kahan), 0.033 (impulse) and 0.0029 (midpoint); the error of the lifted Kahan map grows to 0.19 at $h\omega = 10$. The midpoint rule needed three to four Newton iterations per step, each with a 6×6 linear solve, whereas the lifted Kahan step (Remark 6.4) costs about as much as a Störmer–Verlet step and needs no splitting of the field.

So the lifted Kahan map is an explicit-cost, symmetric, integrable integrator that keeps exact modified energy and angular momenta and invariant tori at step sizes where the standard explicit method fails. Unlike Störmer–Verlet it stays stable beyond $h\omega = 2$, and unlike the impulse method it has a single failure band, whose location is predicted by the sign of $D(z_0)$. It is less accurate than the fully implicit midpoint rule and, between its resonances, than the impulse method, which needs the stiff part to be linear and its flow known. For the exponential and trigonometric class of Theorem 5.1, stiffness that enters through the lifted nonlinear variables limits the step size: for a stiff rotor chain the lifted Kahan method conserved its modified energy and stayed stable up to time 600 for $h\omega \le 2$, and blew up beyond.

9 Conclusions and open problems

Quadratic lifts let Kahan's method act on polynomial, exponential and trigonometric problems. Exact modified integrals survive through a few linear-algebraic mechanisms, collected in Theorem 3.2: affinely moving factors, two-variable integrals, and blocks that move along fixed linear flows at affine speeds, possibly several and non-commuting. These mechanisms protect the lift constraints of quartic potentials, give the Liouville measure for exponential and harmonic potentials, and make the lifted isotropic quartic oscillator integrable in every dimension, with an $O(n)$ explicit form and an explicit solution through a QRT map. For every quartic potential in one degree of freedom the lifted map preserves an explicit measure, integrable or not. Every integrable lifted map we found is covered by Proposition 7.1, including the planar magnetic case, whose integrals lie beyond the block theorem. Open problems:

1. *A block theorem without closure.* The proof of Theorem 6.1(b) uses a block whose closure defect is absorbed by another conserved quantity, and the magnetic integrals of Proposition 7.2 exist although the closure condition fails. A general version of this mechanism, or a reduction theory for Kahan maps with linear symmetries, might derive Proposition 7.2 from Theorem 6.1 through a discrete Larmor theorem, and prove Conjecture 7.3.

2. *Proofs by hand.* Proposition 5.3 and Proposition 7.2(b) are polynomial identities verified by computer algebra in seconds. Hand proofs along the lines of Theorem 6.1(c) would explain the formulas, and might extend Proposition 5.3 to all three-dimensional quadratic Nambu systems, the setting of [20].

3. *Higher order.* Modified integrals depend on h, so compositions of Kahan steps with different step sizes conserve none of them exactly. Is there a lift or a composition that keeps a common invariant?

4. *Applications.* Volume-preserving, reversible, linearly implicit integrators for exponential and rotor lattices (Corollary 5.2) are natural candidates for Hybrid Monte Carlo [39]; their acceptance rates remain to be tested.

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[To be completed.] Parts of the symbolic and numerical computations and of the drafting of this manuscript were carried out with the assistance of an AI system; all results were checked by the author.

Data availability

The scripts that reproduce every computation and figure, including the computer-algebra verifications of Appendix A and the searches of Appendix B, are available at [repository URL to be added].

A Computer-algebra verification

The proofs of Theorem 6.1 in Section 6 are complete without computer assistance. As an independent check, and to complete the proofs of Proposition 5.3, Theorem 6.5 and Proposition 7.2, six SymPy 1.14 scripts in the folder `proofs/` of the repository verify the statements as exact identities between rational functions, with the step size and all parameters symbolic.

1. `quartic_density_proof.py` (under a second) verifies Proposition 5.3: the polynomial form of the measure identity described in its proof, with h and $a_1, \dots, a_4$ symbolic; the values on the sheet; the factorisation for $V = \frac{1}{4}q^4$; and the two expressions through $\sigma_2(F')$ and $\det F'$.

2. `isotropic_proof.py` (about 8 minutes) first checks that the closed form of Remark 6.4 satisfies Kahan's equations identically for $n = 2, 3$, so that it *is* Kahan's map. It then checks the conservation of c_h, M_h and E_h and the measure identity $P_n(z')\det(\mathrm{Id} - \frac{h}{2}F'(z)) = \det(\mathrm{Id} + \frac{h}{2}F'(z'))P_n(z)$ for $n = 2$ and $n = 3$. Since the map and all functions involved are rotation invariant, it suffices to check these on the slices $q = (q_1, 0)$ and $q = (q_1, 0, 0)$, $p = (p_1, p_2, 0)$; the image point is not restricted. Finally it computes a 3×3 minor of the Jacobian of (c_h, M_h, E_h) at $q = (1, 0)$, $p = (0, 1)$, $y = \frac{1}{3}$, which is a non-zero rational function of (h, u_1, u_2); this gives the functional independence used in Corollary 6.2.

3. `isotropic_hand_proof_check.py` verifies each displayed step of the hand proofs, for $n = 3$: (11), the expressions for $q \wedge p'$ and $q' \wedge p$, the relations (12), the two determinants, and the two identities for $2D$ and $2D'$.

4. `isotropic_proof_adjugate.py` computes the step from the adjugate formula $z' = z + h\,\mathrm{adj}(\mathcal{M})F(z)/\det\mathcal{M}$, $\mathcal{M} = \mathrm{Id} - \frac{h}{2}F'(z)$, instead of the closed form, and repeats the checks for $n = 2$. It is slower: the energy takes about 7 minutes, and the measure identity about 10 minutes for $u_2 = \frac{1}{2}$, $u_1 = 0$.

5. `isotropic_radial_check.py` verifies Theorem 6.5. In the variables of its proof, with $|\eta|^2$ and $|\xi|^2$ eliminated as described there, it checks that the numerator of $E_h(z) - e$ is a constant multiple of $D(z)\,\mathcal{Q}(y, y')$; that $E_h = e$ is linear in $|q|^2$ with root $\mathcal{A}(y)$, which gives Q_4; and the limit in part (c). It also checks numerically, along an orbit with $n = 3$ and $u_1 \neq 0$, that $\mathcal{Q}(y_k, y_{k+1}) = 0$ and $b_k^2 = Q_4(y_k)$ to round-off, and that (y_k, b_k) are samples of the elliptic solution at equal steps τ.

6. `magnetic_proof.py` (a few seconds, using sparse polynomial arithmetic) verifies Proposition 7.2 on the slice $q_2 = 0$: the closed form satisfies Kahan's equations; the Larmor identity (15); conservation of c_h, L_h and $\mathcal{E}_h$; the formulas for the two determinants; the measure identity $P^B(z')\det(\mathrm{Id} - \frac{h}{2}F'(z)) = \det(\mathrm{Id} + \frac{h}{2}F'(z'))P^B(z)$; the reductions at $B = 0$ and the limits $h \to 0$; and a non-zero 3×3 minor of the Jacobian of $(c_h, L_h, \mathcal{E}_h)$. The next state is written as a polynomial divided by $Y^2 + \beta^2$, and y' is eliminated at the end through its linear equation.

Independent floating-point and exact rational checks at random points, for $n = 2, 3, 4$, and a floating-point check of Proposition 7.2 that computes the step by a generic linear solve, are also in the repository.

B Exact searches and degree growth

Darboux polynomials and densities. A polynomial P of degree at most d with $P(\Phi_h(z)) = \lambda(z)P(z)$ for a given cofactor λ is found by solving a linear system for its coefficients. We impose the identity at many random points, in arithmetic modulo a large prime p and with a random residue for h, following the discrete-Darboux method of [11]. With the cofactor $\lambda = \det D\Phi_h$ this finds invariant densities $\mathrm{d}z/P$. Since h is a random residue, a solution found this way exists for generic h, and "no solution up to degree d" is an exact negative result for that degree, barring an unlucky choice of prime. All positive results were then confirmed over the rationals or with symbolic h. The modified energy E_h of Theorem 6.1 was found as the ratio of two Darboux polynomials with the cofactor $(K'/K)(Y'/Y)^2$, where K and Y are the factors of the measure.

Parameters in the ansatz. To find the density of Proposition 5.3 for all quartic potentials at once, we treated h and $a_1, \ldots, a_4$ as random residues as well, and sought P as a polynomial in $(h^2, q, p, y, a_1, \ldots, a_4)$ that is invariant under the two scalings of the family of fields, $(q, p, y, h, a_4, a_3, a_2, a_1) \mapsto (\lambda q, \lambda^2 p, \lambda^2 y, \lambda^{-1} h, a_4, \lambda a_3, \lambda^2 a_2, \lambda^3 a_1)$ and $(p, h, a_k) \mapsto (\mu p, \mu^{-1} h, \mu^2 a_k)$. Of degree at most four in (q, p, y) and in h^2 there are 115 such monomials, and the null space is spanned by P and $h^2 a_2 P$ (`code/quartic_density_search.py`).

Invariant bases and exact reconstruction. For maps with a rotation symmetry the search can be restricted to invariant polynomials, which shrinks the linear systems considerably. For the magnetic case (Proposition 7.2) we used the monomials $y^i a^j b^k c^l M^m$ with $a = |q|^2$, $b = q \cdot v$, $c = |v|^2$, $M = q \wedge v$ and $m \leq 1$ (since $M^2 = ac - b^2$), for fixed rational parameters. Solving modulo six primes near 2^{31} and combining the null spaces by the Chinese remainder theorem

and rational reconstruction gives the invariant densities exactly over $\mathbb{Q}$ in under a second (`code/magnetic_darboux_search.py`). For $h = \frac{1}{5}$, $B = 1$, $u_2 = \frac{1}{2}$ and $u_1 = 0$ there is a four-dimensional space of densities of weighted degree at most five; the ratios of its elements give the integrals. Repeating the search for several parameter values identified the coefficients as polynomials in (h, B, u_2), which led to the closed forms of Proposition 7.2, then verified symbolically.

Degree growth. To compute d_k we restrict the k-th iterate to a random line $z = a + bt$ over $\mathbb{F}_p$ and determine the degree of the resulting rational function of t from the nullity of a Padé-type linear system. The iterates are evaluated numerically modulo p rather than symbolically, so degrees of several hundred are within reach. Values above about 400 were not resolved and are reported as "> 400".

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